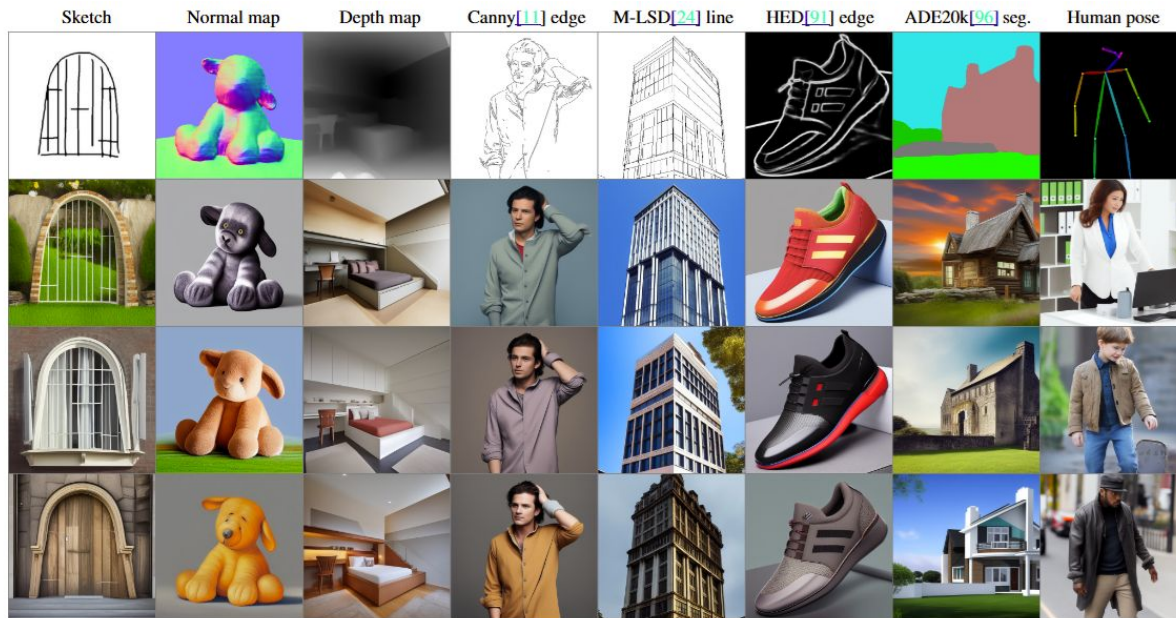
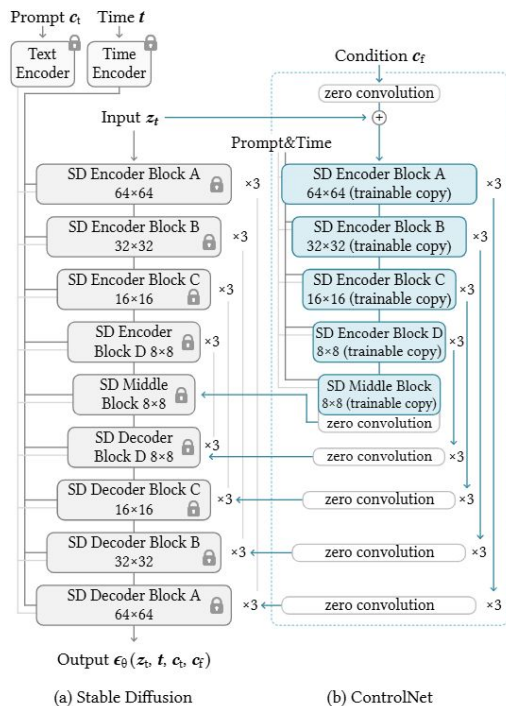


Outline

- Diffusion models
- Diffusion-based Text-to-music Generation
- **Music ControlNet & MuseControlLite**

ControlNet

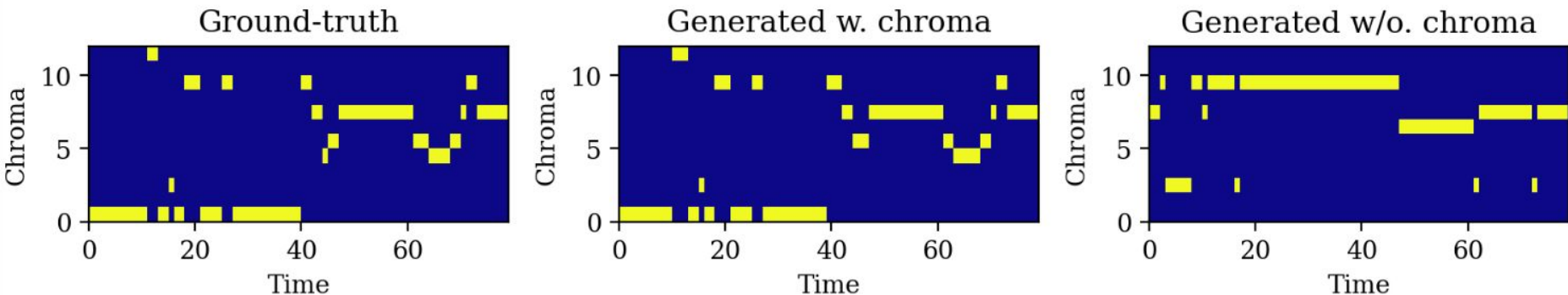


Common Time-varying conditions in Music

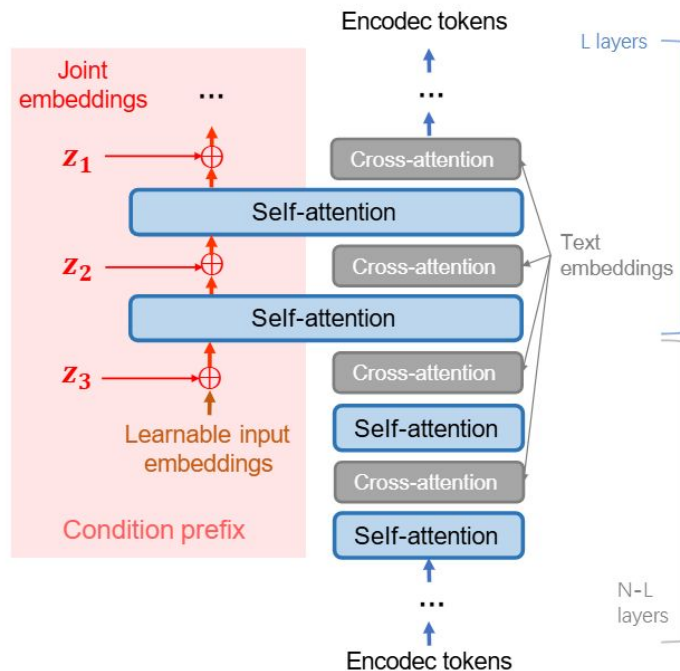
- Melody:
 - STFT-based Chromagram (MusicGen, Music ControlNet)
 - F0 (Jasco)
 - CQT top-4 (Stable-Audio ControlNet, MuseControlLite)
- Rhythm
 - beat time, downbeat time (Music ControlNet, MuseControlLite)
 - Separated drum track (Coco-Mulla)
- Chords
- Dynamics

MusicGen-Melody

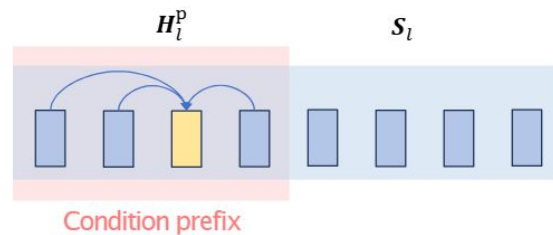
When using melody conditioning, musicgen provide the conditioning tensor C as a prefix to the transformer input.



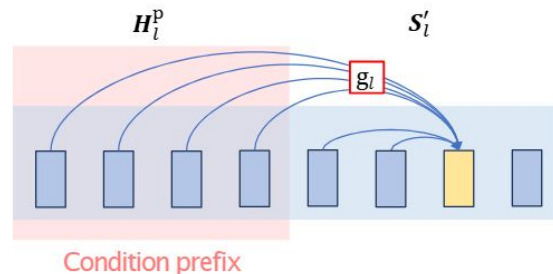
Coco-Mulla



(a) To incorporate content-based control from the joint embeddings, the condition prefix joins into the last L frozen self-attention layers of the MusicGen decoder.



(b) Inside the condition prefix, each position can attend to all other positions within the condition prefix.

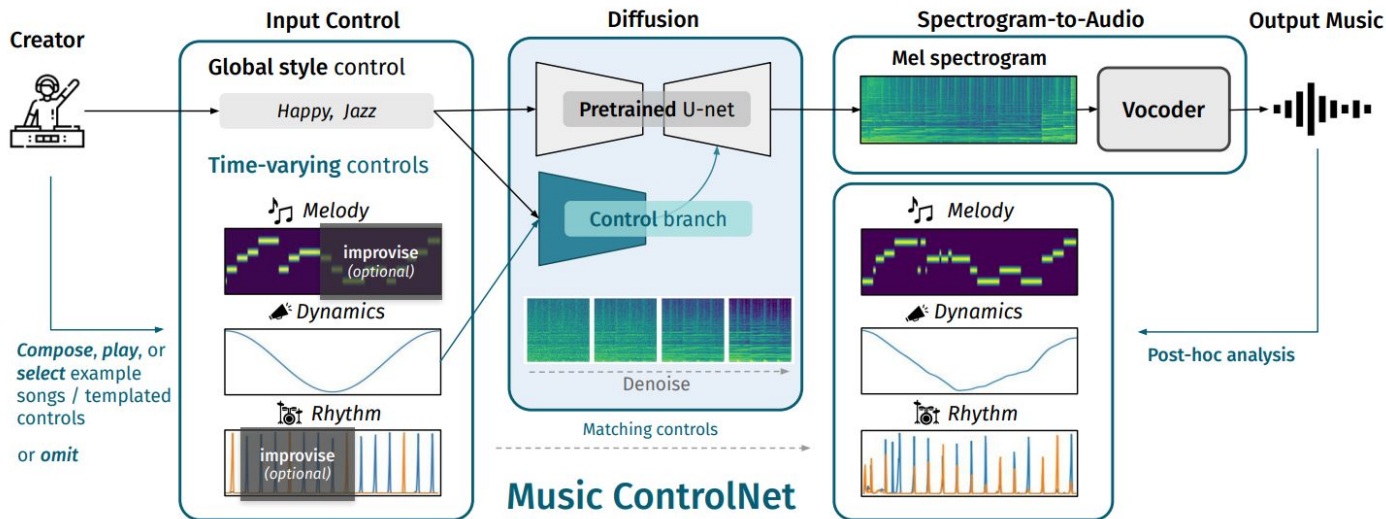


(c) Outside the condition prefix are the hidden embeddings corresponding to Encodec tokens. They attend to all preceding positions (following the causal mask) and the entire condition prefix.

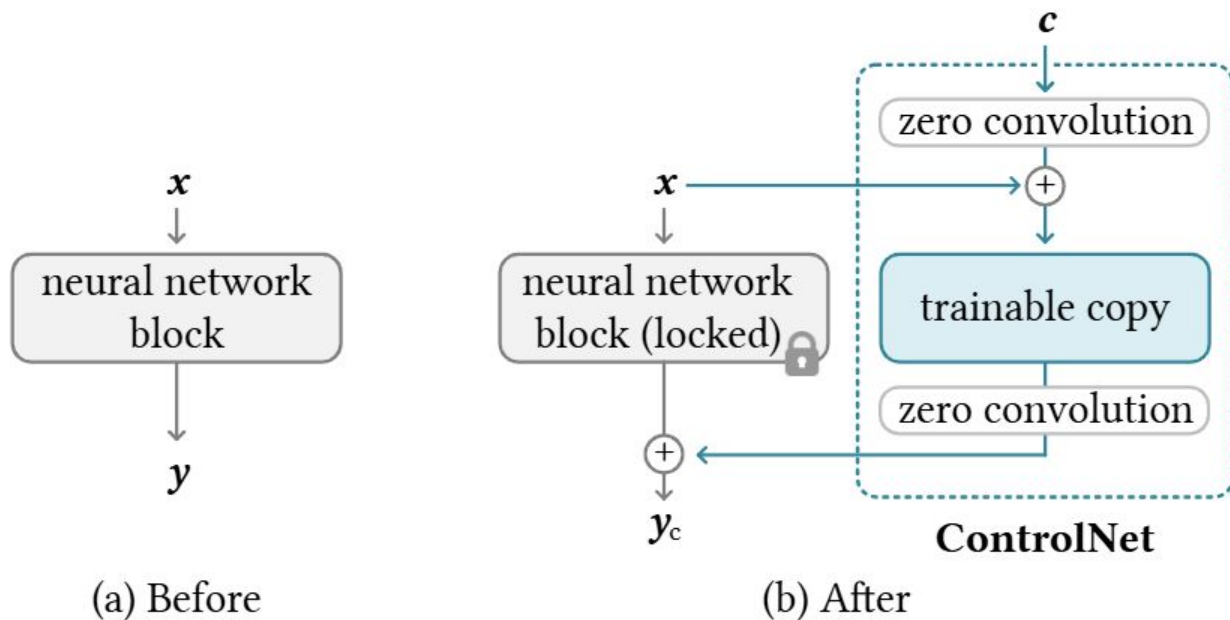
Music ControlNet (for Attribute Control)

<https://musiccontrolnet.github.io/web/>

- Works in the “pixel” domain
- Not open source

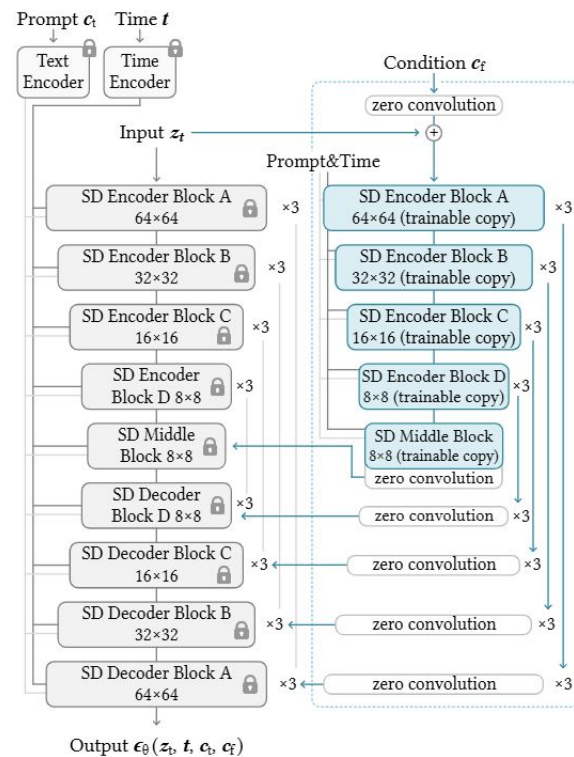


What is ControlNet



What is in the ControlNet Branch

- Each block contains 4 resnet layers and 2 Vision Transformers (ViTs).
- Original branch:
 - Unet: contains 25 blocks
 - Processes: image latent, text
- ControlNet branch:
 - Unet encoder: contains 12 blocks
 - zero convolution layers
 - Processes: condition latents, text



(a) Stable Diffusion

(b) ControlNet

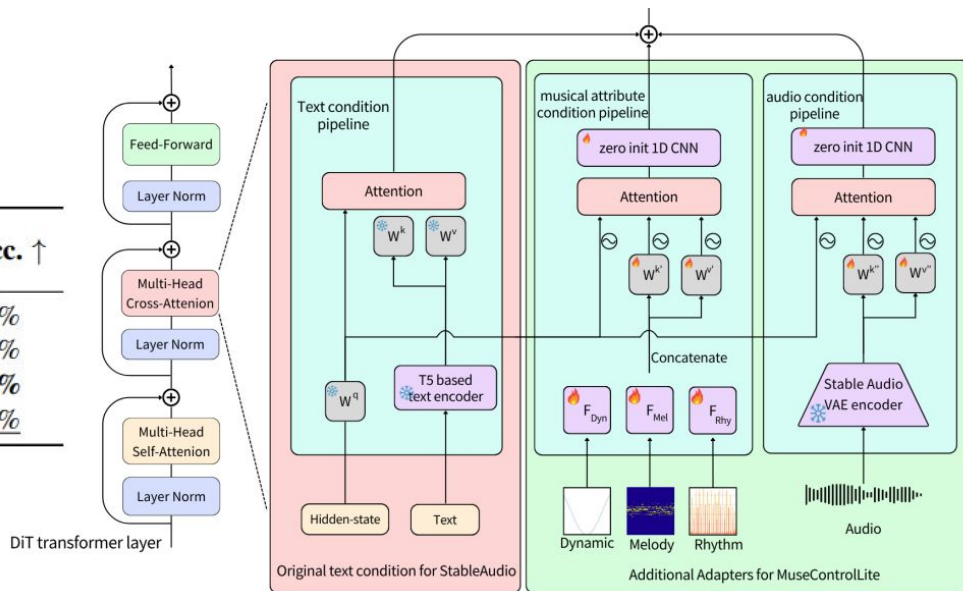
MusicControlLite (for Attribute & Audio Control)

<https://musecontrollite.github.io/web/>

https://colab.research.google.com/drive/1rR-Ncng_gSeb6hX0LY20SA4O9RCF-ZF3

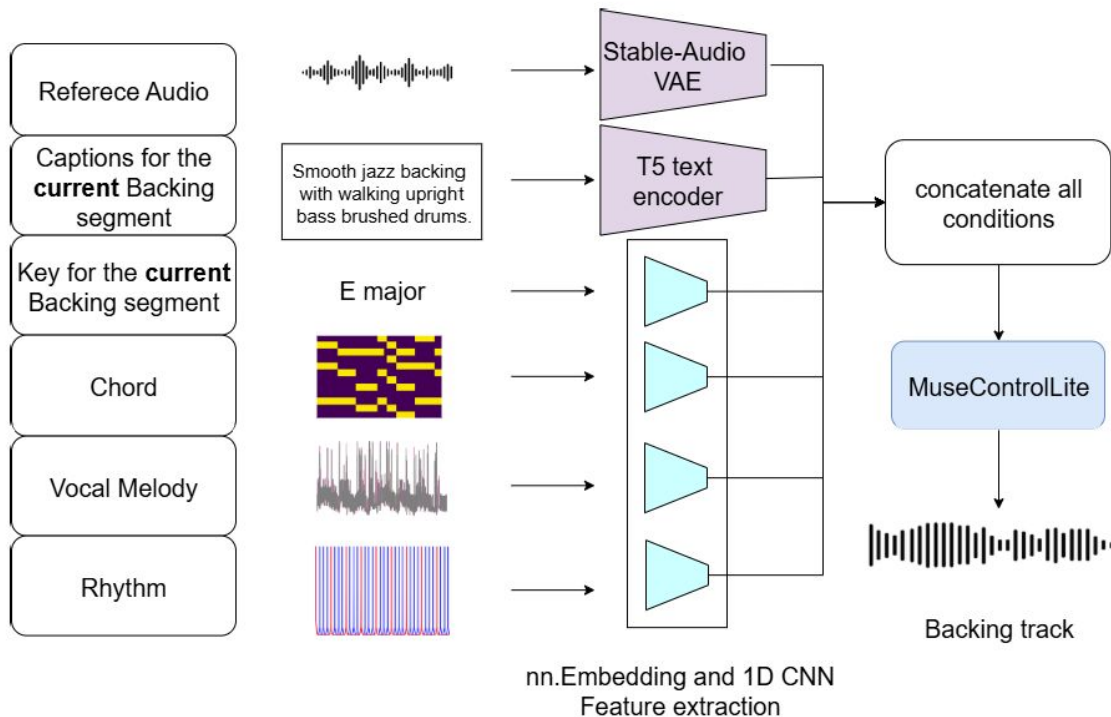
- Latent diffusion
- Open source

Model	FD ↓	KL ↓	CLAP ↑	Mel acc. ↑
MusicGen-Stereo-Large-Melody	193.66	0.436	0.354	43.1%
Stable Audio Open ControlNet	97.73	0.265	0.396	56.6%
Ours (MuseControlLite-Melody)	76.42	0.289	0.372	61.1%
Ours (MuseControlLite-Attr)	<u>80.79</u>	<u>0.271</u>	<u>0.373</u>	<u>60.6%</u>



MusicControlLite (for Singing Accompaniment Generation)

<https://musecontrollite.github.io/web/>



Ref: Tsai et al, "Demonstrating singing accompaniment capabilities for MuseControlLite," AI4music Workshop 2025

Homework-2

Controllable text-to-music generation

HW2 TA: 蔡芳鐸 (Fang-Duo Tsai)
Office hour: Mon. 13:30~15:00 @BL505

Problem Formulation

Generate the **desired** music !

The desired music are in target_music_list_60s.

- 3 piano solo
- 3 drum solo
- 1 Chinese flute solo
- 2 Chineses flute and piano duet



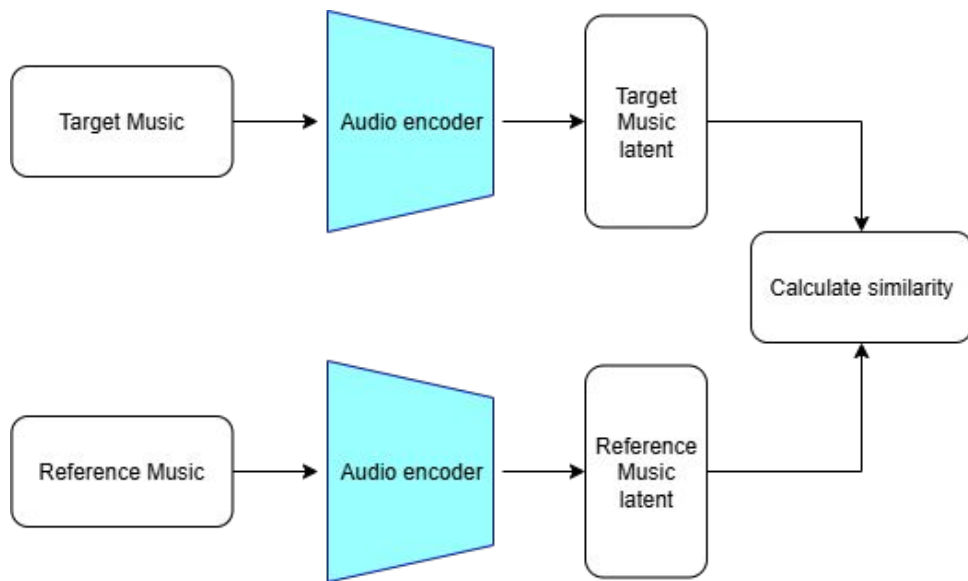
Methods

You are required to implement the following two methods:

1. Retrieval
2. Generation (Controllable text-to-music)

Retrieval

We provide reference_music_list, retrieve the most similar ones for each music in target_music_list.



```
▼ referecne_music_list_60s
1_funk_80_beat_4-4.wav
2_funk_80_beat_4-4.wav
3_funk_80_beat_4-4.wav
04_聽泉.mp3
4_jazz-funk_116_beat_4-4.wav
5_jazz-funk_116_beat_4-4.wav
6_jazz-funk_116_fill_4-4.wav
28_latin-samba_116_beat_4-4.wav
29_latin-samba_116_fill_4-4.wav
30_latin-samba_116_fill_4-4.wav
51_rock-shuffle_125_beat_4-4.wav
76_jazz-mediumfast_180_beat_4-4.wav
77_jazz-mediumfast_180_beat_4-4.wav
78_jazz-fast_290_beat_4-4.wav
79_latin-reggaeton_90_beat_4-4.wav
80_dance-breakbeat_170_beat_4-4.wav
81_dance-breakbeat_170_beat_4-4.wav
```

Suggestions for Audio Encoder

Options include but not limited to:

1. [Stable-Audio-Open VAE encode](#)
2. [Music2latent](#)
3. [CLAP](#)
4. [MuQ](#)

Controllable text-to-music

Method:

1. Audio Captioning
 - a. To obtain the text description of music in target_music_list
2. Text-to-Music
 - a. Simple: Text condition only
 - b. Medium: Any condition extracted from music (e.g. Melody, Rhythm)
 - c. Strong: Adjust the classifier free guidance.

Suggestions for ALMs and Text-to-Music models

Options include but not limited to:

1. ALMs
 - a. [Audio Flamingo 3](#)
 - b. [Owen-audio](#)
 - c. [LP-MusicCaps](#)
2. Controllable Text-to-Music models
 - a. [MuseControlLite](#) (text, rhythm, dynamics, melody)
 - b. [MusicGen](#) (text, melody)
 - c. [MusicGen-style](#) (text, style)
 - d. [Iasco](#) (text, chords, melody, separated drum tracks, full-mix audio)
 - e. [Coco-mulla](#) (pitch, chords, drum track)

The music in target_music_list_60s are 60-second, you can generate music up to the models limit (e.g. MuseControlLite generates 47-second music)

Evaluation metrics (Retrieval)

For each song in the target_music_list, you will have to report:

1. CLAP,
 - a. calculate cosine similarity between:
 - i. Generated music and target music
2. Meta Audiobox Aesthetics
 - a. CE: Content Enjoyment
 - b. CU: Content Usefulness
 - c. PC: Production Complexity
 - d. PQ: Production Quality
3. Melody similarity (accuracy)

* You should slice the target music to the length same as the music your model generates. (i.e. If you use MuseControlLite, trim the target music to the first 47 seconds)

Evaluation metrics (Generation)

For each song in the target_music_list, you will have to report:

1. CLAP,
 - a. calculate cosine similarity between:
 - i. Target music and input text
 - ii. Input text and generated music
 - iii. Generated music and target music
2. Meta Audiobox Aesthetics
 - a. CE: Content Enjoyment
 - b. CU: Content Usefulness
 - c. PC: Production Complexity
 - d. PQ: Production Quality
3. Melody similarity (accuracy)

* You should slice the target music to the length same as the music your model generates.
(i.e. If you use MuseControlLite, trim the target music to the first 47 seconds)

Rules

1. Do the music captioning with ALMs.
2. You can not directly use the music from `target_music_list` as condition to generate music, only condition extracted using MIR tools are allowed.
 - a. You can not use an auto-encoder to encode and decode the target audio, and use it as submission
 - b. You can not use the “audio condition” in MuseControlLite.
3. You can use different methods for each music in `target_music_list`.

Scoring

HW2 accounts for 20% of the total grade

- Report: 100%

Submission

- Report (to NTU cool)
 - For each music in target_music_list, please report
 - Implementation details (e.g. Model you used, time-varying conditions, text input)
 - Generated music (ensure that you set it to public)
 - Evaluation results
 - CLAP
 - Meta Audiobox Aesthetics
 - Melody accuracy
 - Source code url
 - Any thoughts that you will like to share.
- Upload studentID_report.pdf (ex: r12345678_report.pdf)
- Please create a report that is clear and can be understood without the need for oral presentations.

Source code

Upload all your source code and model to a cloud drive, open access permissions, and then put the link in the report. You should provide a detailed readme file, so that I can reproduce the result.

Timeline

- W6 – 10/9 (Thursday): Announcement of HW2
- W9 – 10/29 (Wednesday 23:59pm): Deadline

Late submission: 1 day (-20%), 2 days (-40%), 3 days (-60%), after that (no score)